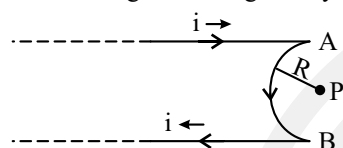


CHAPTER 4

Moving Charges and Magnetism

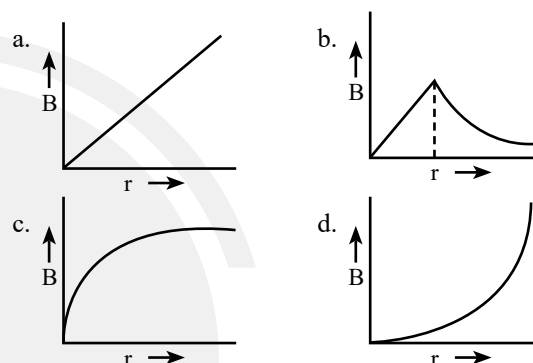
Biot Savart's Law and Magnetic Field Due to Current Through the Straight Conductor

1. A very long conducting wire is bent in a semi-circular shape from A to B as shown in figure. The magnetic field at point P for steady current configuration is given by: (2023)

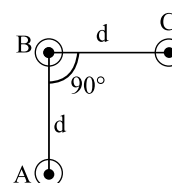


- a. $\frac{\mu_0 i}{4R} \left[1 - \frac{2}{\pi} \right]$ pointed into the page
- b. $\frac{\mu_0 i}{4R}$ pointed into the page
- c. $\frac{\mu_0 i}{4R}$ pointed away from the page
- d. $\frac{\mu_0 i}{4R} \left[1 - \frac{2}{\pi} \right]$ pointed away from page
2. Given below are two statements :
- Statement I :** Biot-Savart's law gives us the expression for the magnetic field strength of an infinitesimal current element ($Id\vec{\ell}$) of the current carrying conductor only.
- Statement II :** Biot-Savart's law is analogous to Coulomb's inverse square law of charge q , with the former being related to the field produced by a scalar source, $Id\vec{\ell}$, while the latter being produced by a vector source, q .
- In light of above statements choose the most appropriate answer from the options given below (2022)
- a. Statement I is incorrect and statement II is correct
- b. Both Statement I and Statement II are correct
- c. Both statement I and Statement II are incorrect
- d. Statement I is correct and Statement II is incorrect.
3. A closely packed coil having 1000 turns has an average radius of 62.8 cm. If current carried by 62.8 cm the wire of the coil is 1 A the value of magnetic field produced at the centre of the coil will be (permeability of free space = $4\pi \times 10^{-7}$ H/m) nearly. (2022 Re)
- a. 10^{-3} T b. 10^{-1} T c. 10^{-2} T d. 10^2 T

4. A thick current carrying cable of radius 'R' carries current 'I' uniformly distributed across its cross-section. The variation of magnetic field $B(r)$ due to the cable with the distance 'r' from the axis of the cable is represented by: (2021)



5. An arrangement of three parallel straight wires placed perpendicular to plane of paper carrying same current 'I' along the same direction is shown in figure. Magnitude of force per unit length on the middle wire 'B' is given by: (2017-Delhi)



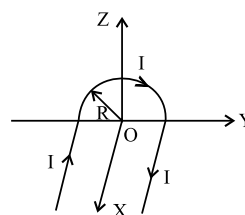
- a. $\frac{2\mu_0 i^2}{\pi d}$ b. $\frac{\sqrt{2}\mu_0 i^2}{\pi d}$
- c. $\frac{\mu_0 i^2}{\sqrt{2}\pi d}$ d. $\frac{\mu_0 i^2}{2\pi d}$
6. Two identical long conducting wires AOB and COD are placed at right angle to each other, with one above other such that O is their common point for the two. The wires carry I_1 and I_2 currents, respectively. Point P is lying at distance d from O along a direction perpendicular to the plane containing the wires. The magnetic field at the point P will be: (2014)
- a. $\frac{\mu_0}{2\pi d} \left(\frac{I_1}{I_2} \right)$ b. $\frac{\mu_0}{2\pi d} (I_1 + I_2)$
- c. $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)$ d. $\frac{\mu_0}{2\pi d} (I_1^2 + I_2^2)^{1/2}$

Magnetic Field in Circular Coil

7. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take permeability of free space as $4\pi \times 10^{-7}$ SI units): (2024)
- a. 4.4 mT b. 44 T c. 44 mT d. 4.4 T
8. A long wire carrying a steady current is bent into a circular loop of one turn. The magnetic field at the center of the loop is B. It is then bent into a circular coil of n turns. The magnetic field at the center of this coil of n turns will be: (2016 - II)
- a. 2nB b. $2n^2B$ c. nB d. n^2B
9. An electron moving in a circular orbit of radius r makes n rotations per second. The magnetic field produced at the center has magnitude: (2015)
- a. Zero b. $\frac{\mu_0 n^2 e}{r}$ c. $\frac{\mu_0 n e}{2r}$ d. $\frac{\mu_0 n e}{2\pi r}$

Magnetic Field Due to Current Through Both Straight Wire and Circular Wire (Mixed Figure)

10. A wire carrying a current I along the positive x-axis has length L. It is kept in a magnetic field $\vec{B} = (2\hat{i} + 3\hat{j} - 4\hat{k})$ T. The magnitude of the magnetic force acting on the wire is: (2023)
- a. $\sqrt{3} IL$ b. 3 IL c. $\sqrt{5} IL$ d. 5 IL
11. A long straight wire of length 2 m and mass 250 g is suspended horizontally in a uniform horizontal magnetic field of 0.7 T. The amount of current flowing through the wire will be ($g = 9.8 \text{ ms}^{-2}$): (2023-Manipur)
- a. 2.45 A b. 2.25 A c. 2.75 A d. 1.75 A
12. The shape of the magnetic field lines due to an infinite long, straight current carrying conductor is: (2022 Re)
- a. a plane b. a straight line
c. circular d. elliptical
13. Two very long, straight, parallel conductors A and B carry current of 5 A and 10 A respectively and are at a distance of 10 cm from each other. The direction of current in two conductors is same. The force acting per unit length between two conductors is: (2022 Re)
- ($\mu_0 = 4\pi \times 10^{-7}$ SI unit)
- a. $1 \times 10^{-4} \text{ Nm}^{-1}$ and is repulsive
b. $2 \times 10^{-4} \text{ Nm}^{-1}$ and is attractive
c. $2 \times 10^{-4} \text{ Nm}^{-1}$ and is repulsive
d. $1 \times 10^{-4} \text{ Nm}^{-1}$ and is attractive
14. The magnetic field on the axis of a circular loop of radius 100 cm carrying current $I = \sqrt{2}$ A, at point 1 m away from the centre of the loop is given by: (2022 Re)
- a. 6.28×10^{-4} T b. 3.14×10^{-7} T
c. 6.28×10^{-7} T d. 3.14×10^{-4} T
15. A wire carrying current I has the shape as shown in adjoining figure. Linear parts of the wire are very long and parallel to X-axis while semicircular portion of radius R is lying in Y-Z plane. Magnetic field at point O is: (2015)



- a. $\vec{B} = -\frac{\mu_0 I}{4\pi R}(\pi\hat{i} - 2\hat{k})$ b. $\vec{B} = -\frac{\mu_0 I}{4\pi R}(\pi\hat{i} + 2\hat{k})$
c. $\vec{B} = \frac{\mu_0 I}{4\pi R}(\pi\hat{i} - 2\hat{k})$ d. $\vec{B} = \frac{\mu_0 I}{4\pi R}(\pi\hat{i} + 2\hat{k})$

Ampere's Circuital Law

16. From Ampere's circuital law for a long straight wire of circular cross section carrying a steady current, the variation of magnetic field in the inside and outside region of the wire is: (2022)
- a. a linearly decreasing function of distance upto the boundary of the wire and then a linearly increasing one for the outside region.
b. uniform and remains constant for both the regions.
c. a linearly increasing function of distance upto the boundary of the wire and then decreasing outside region.
d. a linearly increasing function of distance r upto the boundary of the wire $1/r$ dependence for the outside region.
17. A cylindrical conductor of radius R is carrying a constant current. The plot of the magnitude of the magnetic field B with the distance d from the centre of the conductor, is correctly represented by the figure: (2019)
- a. b.
c. d.
18. A long straight wire of radius a carries a steady current I. The current is uniformly distributed over its cross-section. The ratio of the magnetic fields B and B' at radial distances $\frac{a}{2}$ and 2a respectively, from the axis of the wire is: (2016 - I)

- a. $\frac{1}{4}$ b. $\frac{1}{2}$ c. 1 d. 4

Magnetic Field Due to Solenoid and Toroid

19. A long solenoid of radius 1 mm has 100 turns per mm. If 1 A current flows in the solenoid, the magnetic field strength at the centre of the solenoid is: (2022)
- a. 6.28×10^{-4} T b. 6.28×10^{-2} T
c. 12.56×10^{-2} T d. 12.56×10^{-4} T

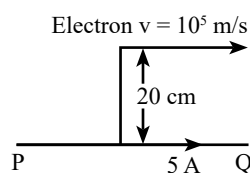
20. A long solenoid of 50 cm length having 100 turns carries a current of 2.5 A. The magnetic field at the centre of the solenoid is : (2020)

$$(\mu_0 = 4\pi \times 10^{-7} \text{ TmA}^{-1})$$

- a. $3.14 \times 10^{-4} \text{ T}$ b. $6.28 \times 10^{-5} \text{ T}$
c. $3.14 \times 10^{-5} \text{ T}$ d. $6.28 \times 10^{-4} \text{ T}$

Motion of Charged Particle in (i) Electric Field and (ii) Magnetic Field

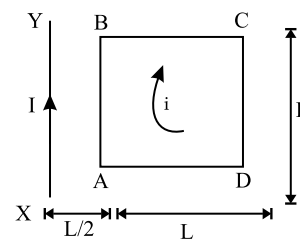
21. An infinitely long straight conductor carries a current of 5 A as shown. An electron is moving with a speed of 10^5 m/s parallel to the conductor. The perpendicular distance between the electron and the conductor is 20 cm at an instant. Calculate the magnitude of the force experienced by the electron at that instant. (2021)



- a. $8\pi \times 10^{-20} \text{ N}$ b. $4\pi \times 10^{-20} \text{ N}$
c. $8 \times 10^{-20} \text{ N}$ d. $4 \times 10^{-20} \text{ N}$
22. Ionized hydrogen atoms and α -particles with same momenta enters perpendicular to a constant magnetic field, B. The ratio of their radii of their paths $r_H : r_\alpha$ will be : (2019)
a. 2 : 1 b. 1 : 2 c. 4 : 1 d. 1 : 4
23. An electron is moving in a circular path under the influence of a transverse magnetic field of $3.57 \times 10^{-2} \text{ T}$. If the value of e/m is $1.76 \times 10^{11} \text{ C/kg}$, the frequency of revolution of the electron is: (2016 - II)
a. 62.8 MHz b. 6.28 MHz
c. 1 GHz d. 100 MHz
24. A proton and an alpha particle both enter a region of uniform magnetic field, B, moving at right angles to the field B. If the radius of circular orbits for both the particles is equal and the kinetic energy acquired by proton is 1 MeV, the energy acquired by the alpha particle will be: (2015 Pre)
a. 1 MeV b. 4 MeV c. 0.5 MeV d. 1.5 MeV

Force in the Presence of Magnetic Field, Lorentz Force and Cyclotron

25. A uniform electric field and a uniform magnetic field are acting along the same direction in a certain region. If an electron is projected in the region such that its velocity is pointed along the direction of fields, then the electron: (2023-Manipur)
a. will turn towards right of direction of motion
b. will turn towards left of direction of motion
c. speed will decrease
d. speed will increase
26. A square loop ABCD carrying a current i , is placed near and coplanar with a long straight conductor XY carrying a current I , the net force on the loop will be: (2016 - I)



- a. $\frac{2\mu_0 Ii}{3\pi}$ b. $\frac{\mu_0 Ii}{2\pi}$ c. $\frac{2\mu_0 IiL}{3\pi}$ d. $\frac{\mu_0 IiL}{2\pi}$

27. When a proton is released from rest in a room, it starts with an initial acceleration a_0 towards west. When it is projected towards north with a speed v_0 it moves with an initial acceleration $3a_0$ toward west. The electric and magnetic fields in the room are: (2013)

- a. $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ down b. $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ up
c. $\frac{ma_0}{e}$ west, $\frac{2ma_0}{ev_0}$ down d. $\frac{ma_0}{e}$ east, $\frac{3ma_0}{ev_0}$ up

Torque on a Current Loop

28. A rectangular coil of length 0.12 m and width 0.1 m having 50 turns of wire is suspended vertically in a uniform magnetic field of strength 0.2 Weber/m². The coil carries a current of 2 A. If the plane of the coil is inclined at an angle of 30° with the direction of the field, the torque required to keep the coil in stable equilibrium will be: (2015 Pre)
a. 0.12 Nm b. 0.15 Nm c. 0.20 Nm d. 0.24 Nm
29. A current loop in a magnetic field: (2013)
a. Can be in equilibrium in two orientations, one stable while the other is unstable
b. Experiences a torque whether the field is uniform or non uniform in all orientations
c. Can be in equilibrium in one orientation
d. Can be in equilibrium in two orientations, both the equilibrium states are unstable

Galvanometer

30. Current sensitivity of a moving coil galvanometer is 5 div/mA and its voltage sensitivity (angular deflection per unit voltage applied) is 20 div/V. The resistance of the galvanometer is : (2018)
a. 250 Ω b. 25 Ω c. 40 Ω d. 500 Ω

Conversion of Galvanometer into (a) Ammeter and (b) Voltmeter

31. In an ammeter 0.2% of main current passes through the galvanometer. If resistance of galvanometer is G , the resistance of ammeter will be: (2014)
a. $\frac{1}{499}G$ b. $\frac{499}{500}G$ c. $\frac{1}{500}G$ d. $\frac{500}{499}G$

Magnetic Moment

32. An iron bar of length L has magnetic moment M . It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is: (2024)

- a. $2M$ b. $\frac{M}{\sqrt{3}}$ c. M d. $\frac{M}{2}$

33. A uniform conducting wire of length $12a$ and resistance ' R ' is wound up as a current carrying coil in the shape of,

- i. an equilateral triangle of side ' a '.
ii. a square of side ' a '.

The magnetic dipole moments of the coil in each case respectively are: (2021)

- a. $3Ia^2$ and Ia^2 b. $3Ia^2$ and $4Ia^2$
c. $4Ia^2$ and $3Ia^2$ d. $\sqrt{3}Ia^2$ and $3Ia^2$

34. A wire of length L metre carrying a current of I ampere is bent in the form of a circle. its magnetic moment is. (2020-Covid)

- a. $I\pi L^2 / 4 \text{ Am}^2$ b. $2I L^2 / \pi \text{ Am}^2$
c. $I L^2 / 4 \pi \text{ Am}^2$ d. $I L^2 / 4 \text{ Am}^2$

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (c) | 4. (b) | 5. (c) | 6. (d) | 7. (a) | 8. (d) | 9. (c) | 10. (d) |
| 11. (d) | 12. (c) | 13. (d) | 14. (b) | 15. (b) | 16. (d) | 17. (c) | 18. (c) | 19. (c) | 20. (d) |
| 21. (c) | 22. (a) | 23. (c) | 24. (a) | 25. (c) | 26. (a) | 27. (c) | 28. (c) | 29. (a) | 30. (a) |
| 31. (c) | 32. (d) | 33. (d) | 34. (c) | | | | | | |